Kirchhoff plate report
Vygenerováno: 2026-03-09T10:46:54

Zadání:

```
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  "q": 15.0,
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  "nu": 0.25,
  "n_query_pts": 100,
  "edges_text": "0.0,0.0,1,1\n4.0,0.0,0,0\n4.0,2.0,0,0\n3.0,2.0,0,0\n2.0,3.0,0,0\n2.0,5.0,0,0\n3.0,6.0,0,0\n4.0,6.0,0,0\n4.0,8.0,1,1\n0.0,8.0,1,1",
  "line_par_x": [
    [
      0.0,
      4.0,
      0.5
    ],
    [
      0.0,
      4.0,
      1.5
    ],
    [
      0.0,
      2.5,
      2.5
    ]
  ],
  "line_par_y": [
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    8.0,
    1.9
  ],
  "field_extrema": {
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        "x": 0.029044942468614117,
        "y": 3.950189578402116
      },
      "max": {
        "value": 7.982617907473191,
        "x": 3.0018331018202535,
        "y": 6.056401860626262
      }
    },
    "my": {
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        "x": 3.749432100082485,
        "y": 0.024869132586356417
      },
      "max": {
        "value": 8.82649042962146,
        "x": 1.9429338924866344,
        "y": 4.998406481890957
      }
    },
    "mxy": {
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        "value": -5.598824530645224,
        "x": 1.9686782352259797,
        "y": 2.950971767900878
      },
      "max": {
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        "x": 1.9685374059469567,
        "y": 5.0489374016029664
      }
    },
    "mx_dim_lower": {
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        "value": -30.264105041362907,
        "x": 0.02914019882018934,
        "y": 4.050217129967346
      },
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        "value": 12.68720239400211,
        "x": 3.0016747038044707,
        "y": 1.943537664228517
      }
    },
    "my_dim_lower": {
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        "x": 3.6493458269660284,
        "y": 7.973202476159785
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        "y": 3.950189578402116
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    },
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    }
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}
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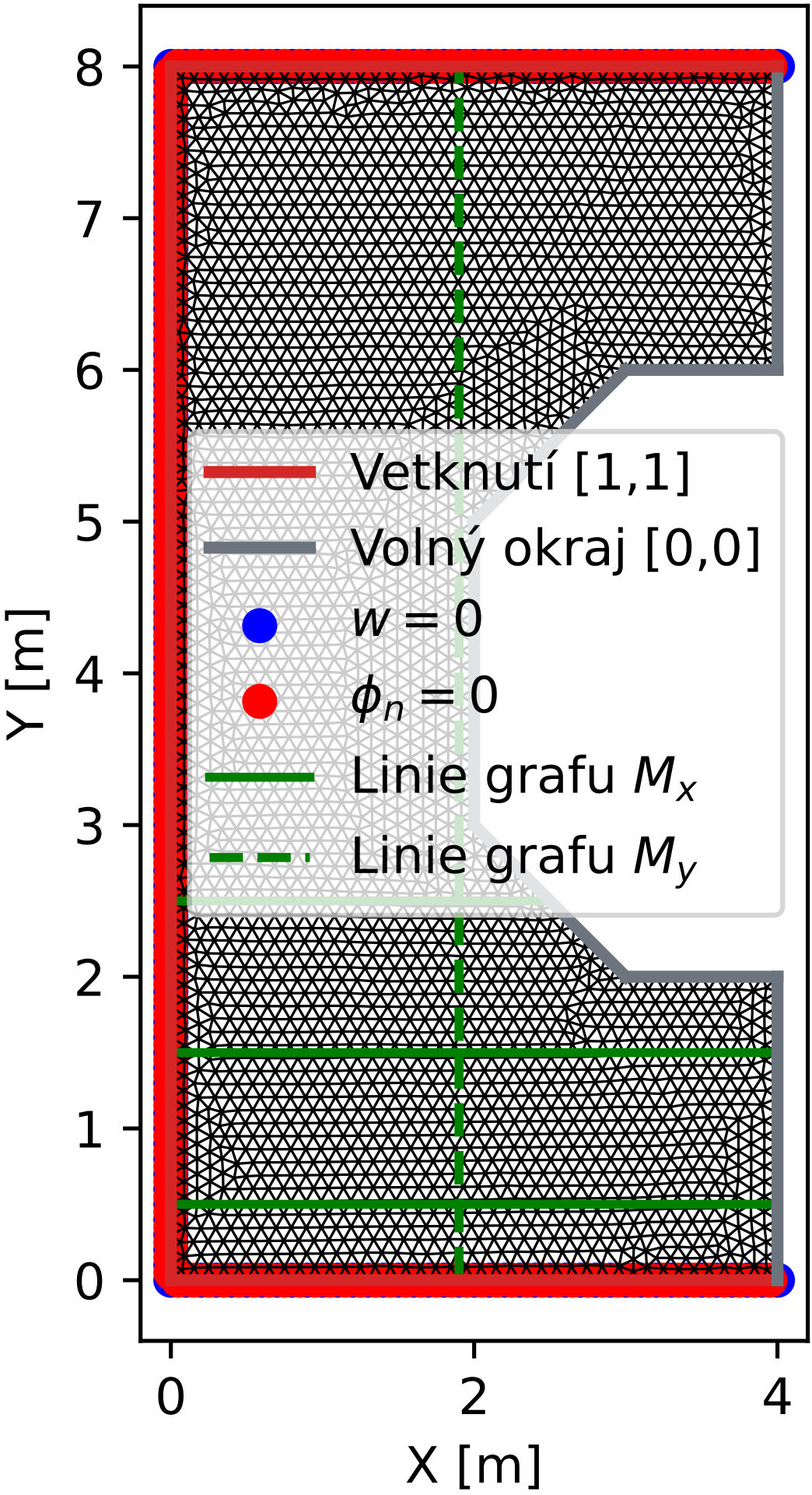
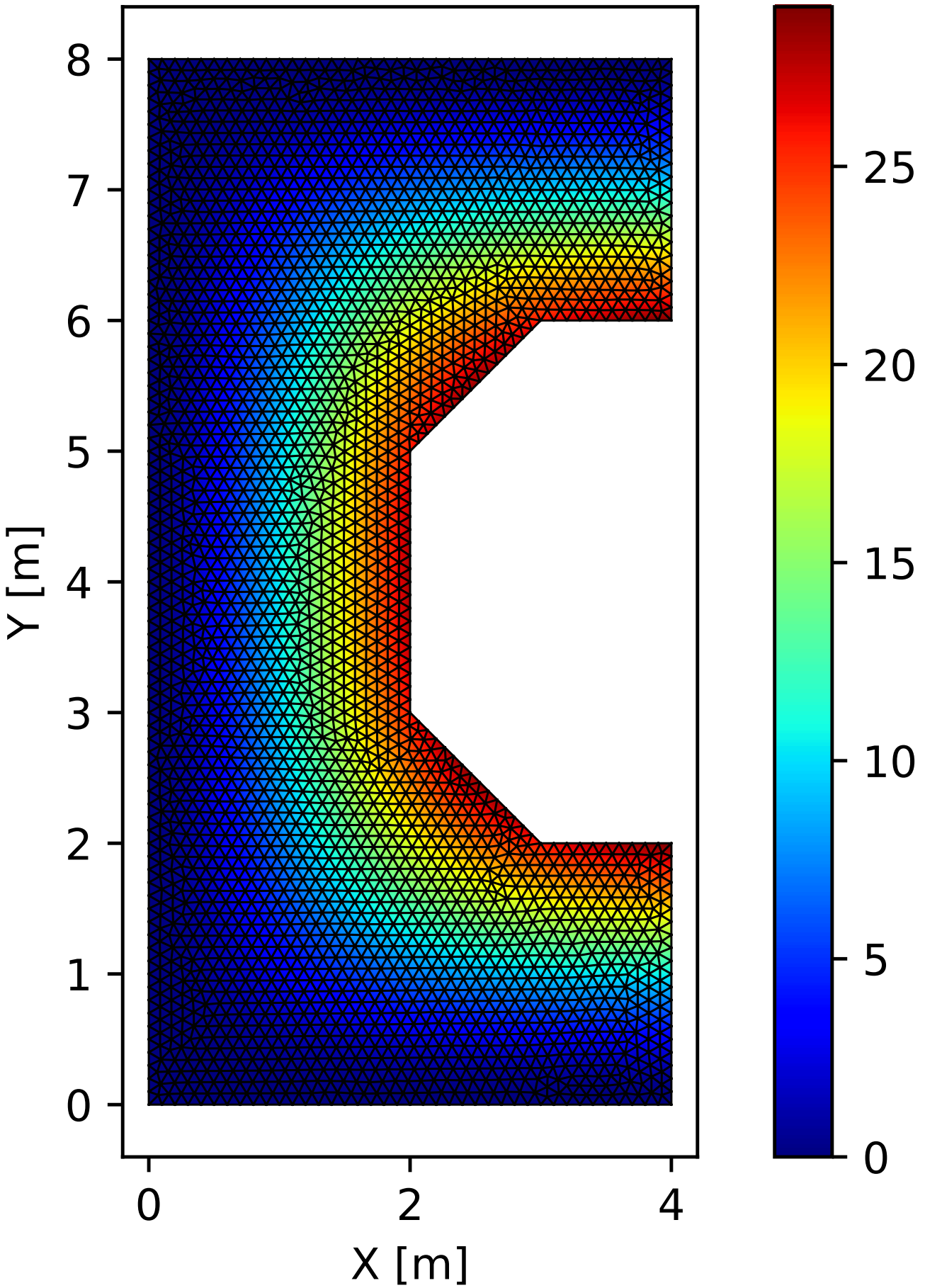
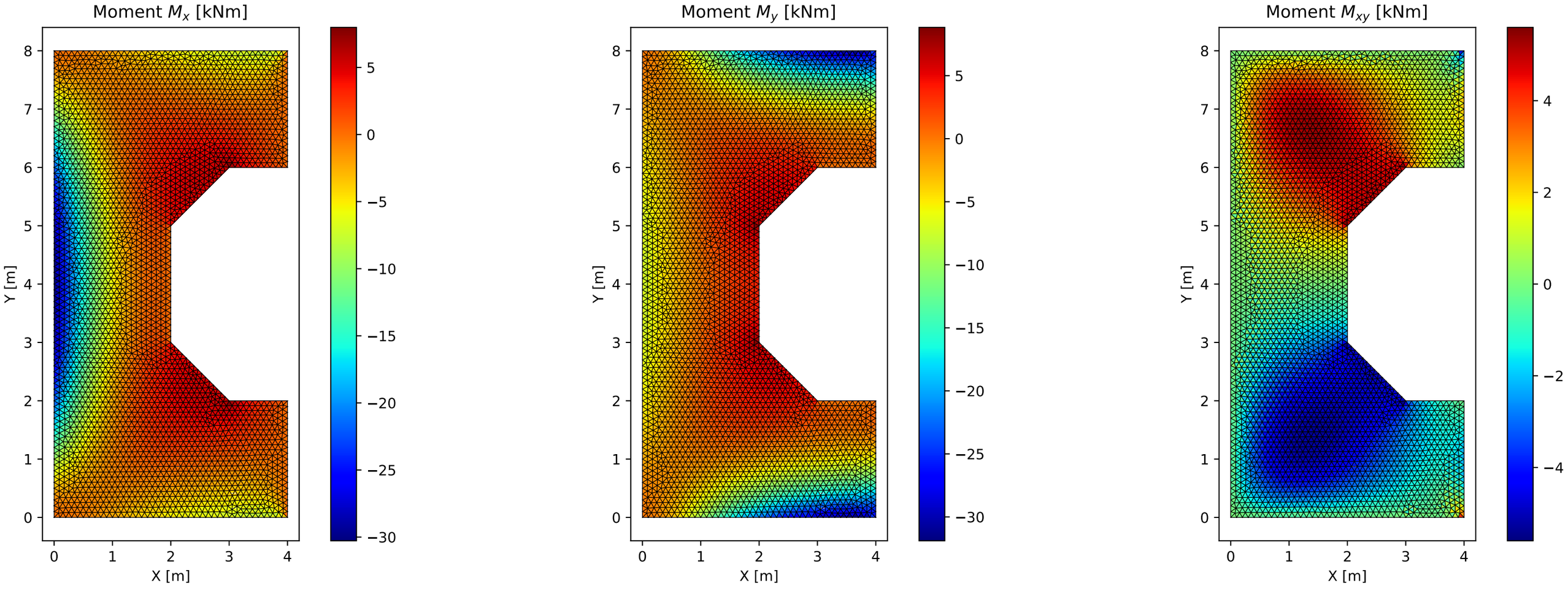
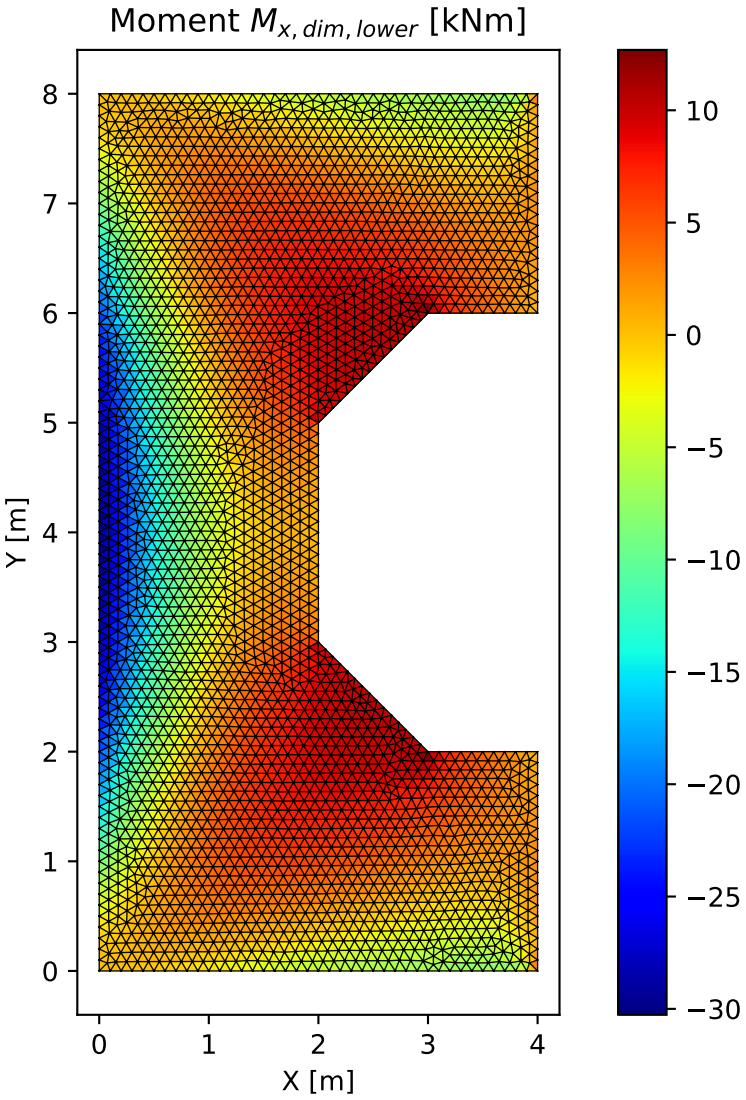
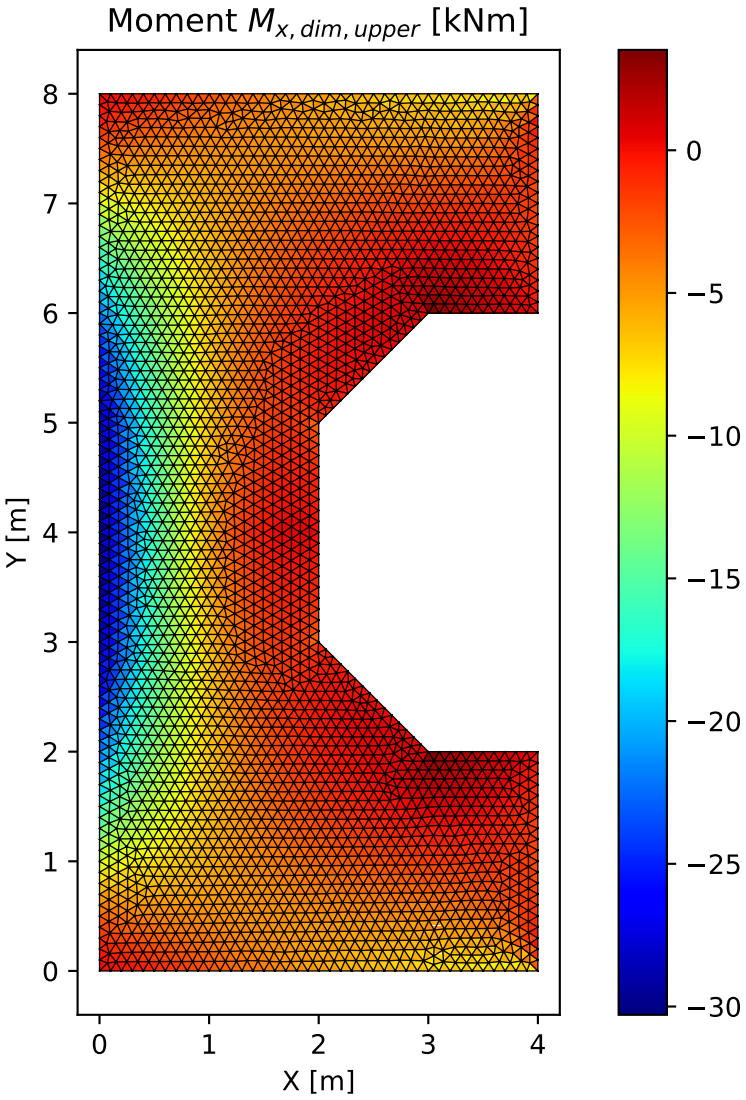

Vetknutí [1,1]
Volný okraj [0,0]
$w = 0$
$\phi_n = 0$
Linie grafu M_x
Linie grafu M_y
Y [m]
X [m]
0
1
2
3
4
5
6
7
8
0
2
4

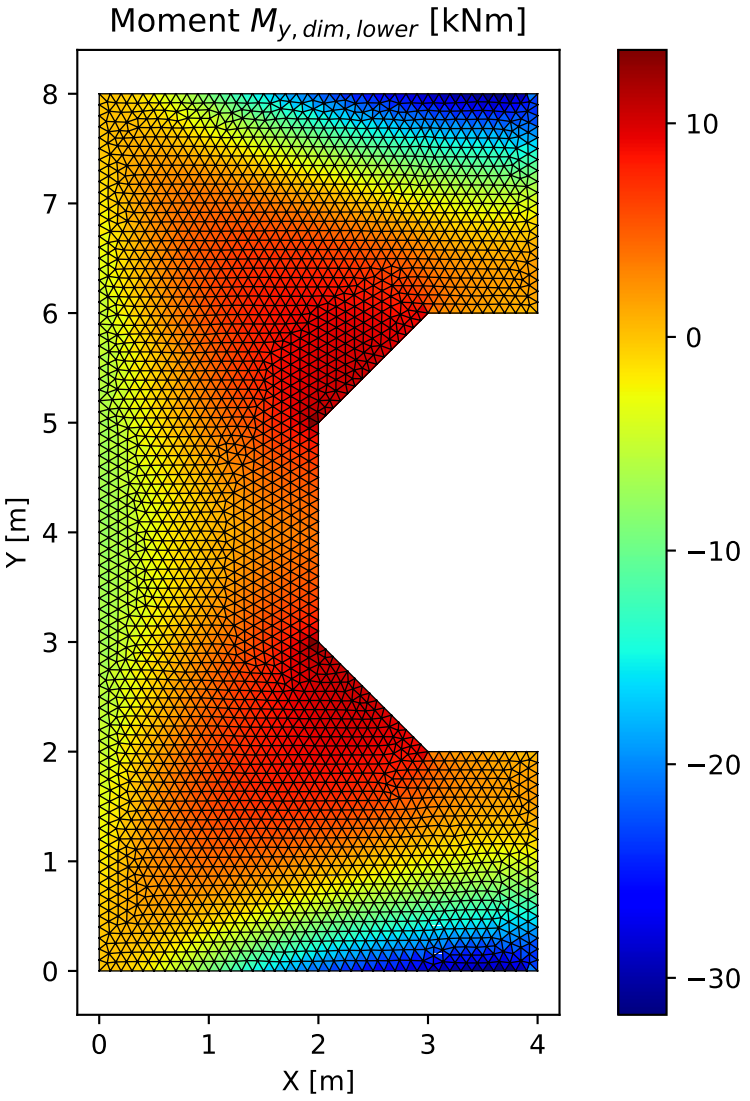
Průhyb $w(x, y)$ pro deskovou tuhost D = 1 kNm

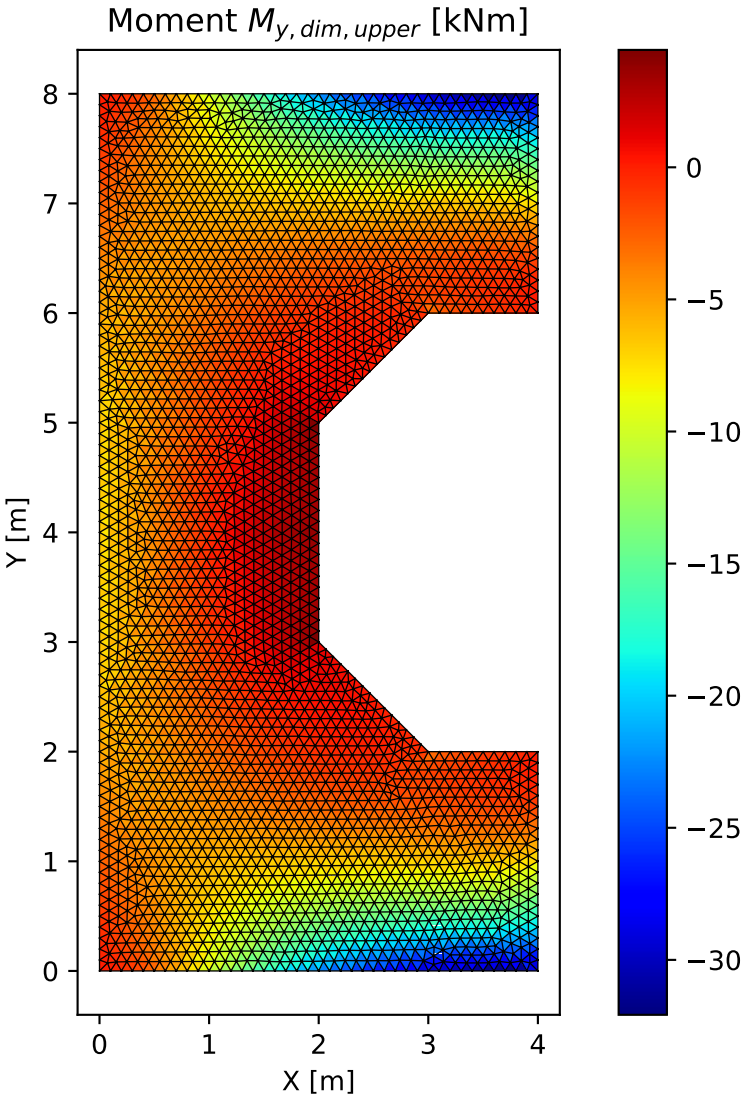


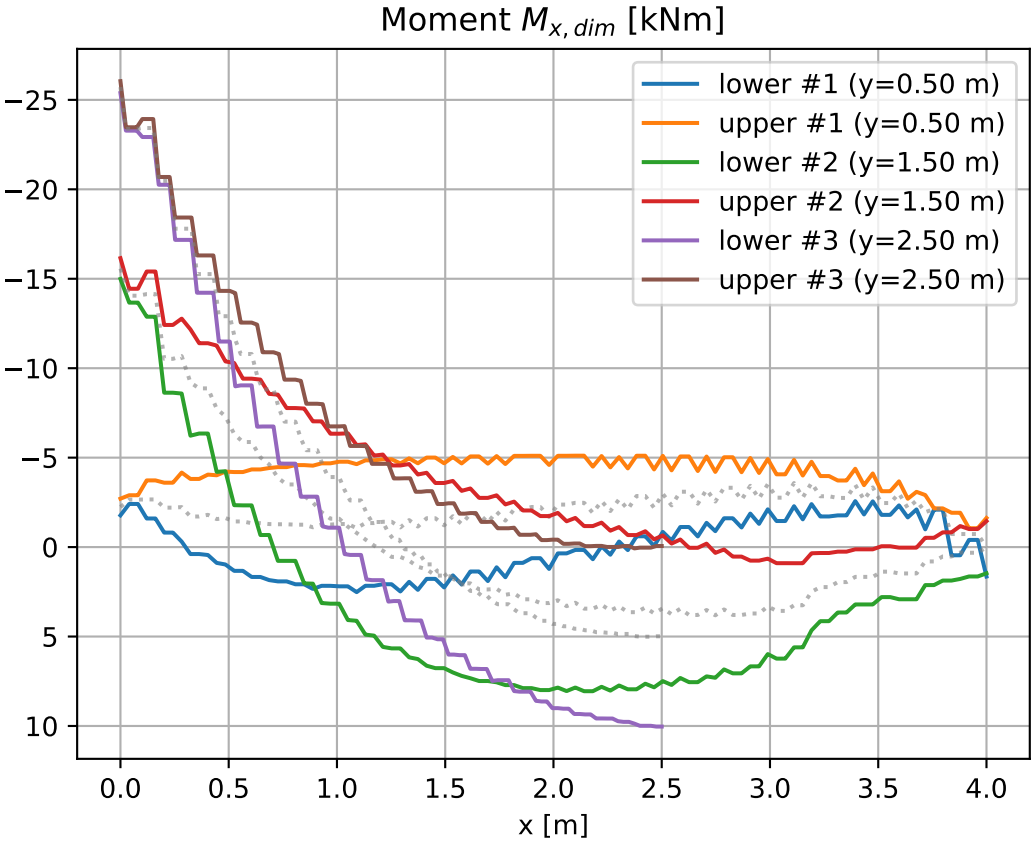

Moment M_x [kNm]
Moment M_y [kNm]
Moment M_{xy} [kNm]
X [m]
Y [m]
0
1
2
3
4
5
6
7
8
5
0
−5
−10
−15
−20
−25
−30
4
2
0
−2
−4

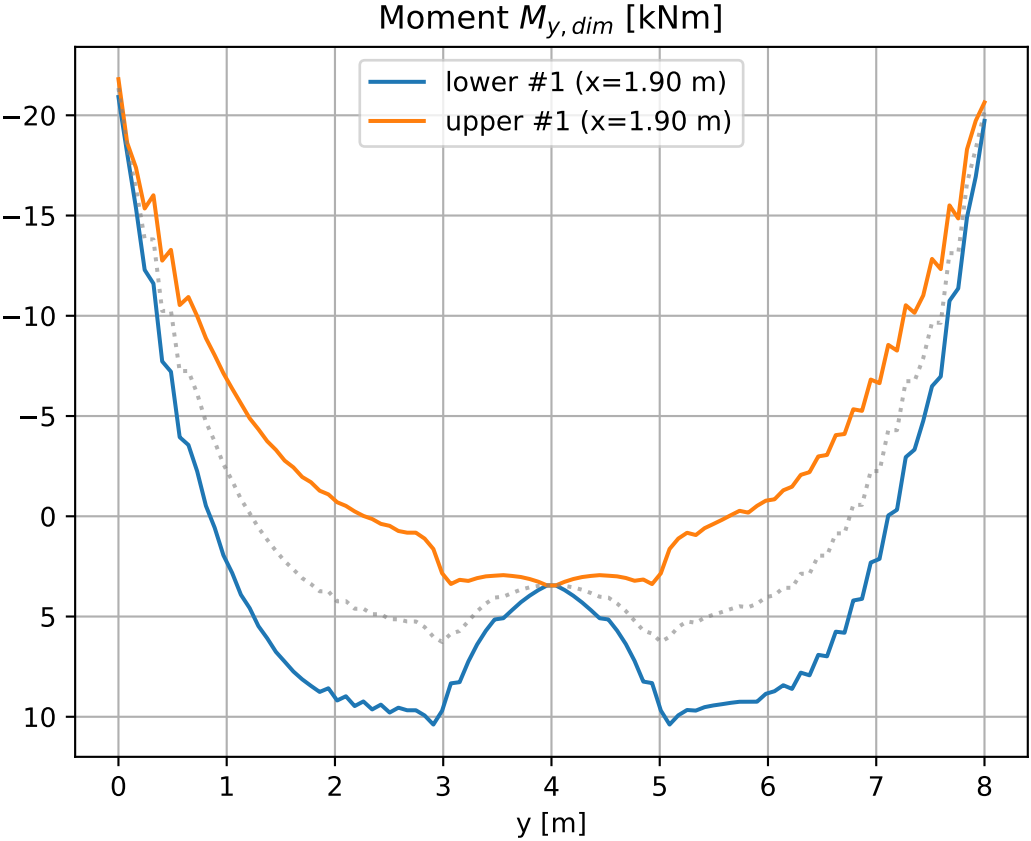

Moment $M_{x,dim,lower}$ [kNm]
Y [m]
X [m]
0
1
2
3
4
5
6
7
8
10
5
0
−5
−10
−15
−20
−25
−30


Moment $M_{x,dim,upper}$ [kNm]
Y [m]
X [m]
0
1
2
3
4
5
6
7
8
0
−5
−10
−15
−20
−25
−30


Moment $M_{y,dim,lower}$ [kNm]
Y [m]
X [m]
0
1
2
3
4
5
6
7
8
10
0
−10
−20
−30


Moment $M_{y,dim,upper}$ [kNm]
Y [m]
X [m]
0
1
2
3
4
5
6
7
8
0
−5
−10
−15
−20
−25
−30


Moment $M_{x,dim}$ [kNm]
lower #1 (y=0.50 m)
upper #1 (y=0.50 m)
lower #2 (y=1.50 m)
upper #2 (y=1.50 m)
lower #3 (y=2.50 m)
upper #3 (y=2.50 m)
−25
−20
−15
−10
−5
0
5
10
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
x [m]


Moment $M_{y,dim}$ [kNm]
lower #1 (x=1.90 m)
upper #1 (x=1.90 m)
−20
−15
−10
−5
0
5
10
0
1
2
3
4
5
6
7
8
y [m]